Fleet Scheduling, Admission, and Failure Recovery

Deep Dive into Kimi K3 — Chapter 45

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Goal: follow six sessions on C0 through a cluster failure — price their remaining work, place them across failure domains, admit them under budgets, fence old authority, and replay without a second overload.

Roadmap

Pricing and placing work

Admission, fairness, and authority

Elasticity, recovery, and evidence

Pricing and placing work

Remaining work, affinity, and failure domains

- Demand is a three-axis vector (prefill, decode, transfer MiB), never summed: A warm $(10, 4, 0)$, cold $(100, 4, 0)$; memory obeys $M_j + U_j + \delta \leq B_j$.

Remaining work, affinity, and failure domains

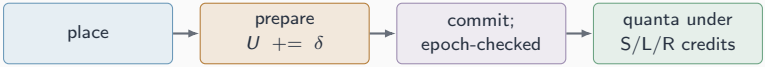
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- Rendezvous ranking gives distinct primary/secondary domains; counts $(2, 2, 2)$ still hide replay work $(110, 70, 50)$, max/mean 33/23.

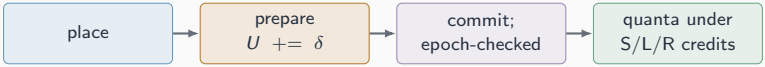
Admission, fairness, and authority

Budgets, lower service, and one owner



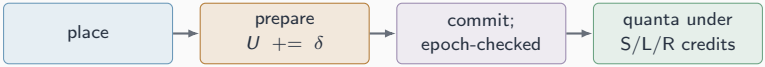
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- Retiring grants stay charged until quiescent: fencing stops publication, not an issued device write.

Elasticity, recovery, and evidence

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Restoring every lost session at once can overload the survivors. Next: Chapter 46's Low-Precision Execution and Speculative Decoding.